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**Date:** 2/27/2026

**Class:** Engineer Design

## Element A: Presentation and Justification of the Problem

### Problem Statement and Details (A1, A2, A5)

The A.C.T. dust collection system in the Walpole High School woodworking shop is currently underperforming due to an open port in one of the two extension ducts over the work tables (see Figure 1). Installed several years ago to ensure breathable air and cleaner work environment for students and staff when machines are running, the system's inefficiency poses a potential health risk if not addressed. As A.C.T.'s website states, "a properly functioning system is vital for improving employee efficiency and helping to ensure a safe working environment." (A.T.C pdf)

Our team is tasked with solving an efficiency problem with the dust collection system in the Walpole High School woodshop. Currently, when the metal vent is open and one hose is connected, an open port, as shown in Figures 1 and 2, reduces the system's efficiency. This open port needs a closure of some sort to improve the vacuum system.

The core issue is that the open port must be blocked to prevent air flow when other machines are running. This problem becomes apparent when the system is on and machines are cutting or sanding wood which creates wood and dust particles throughout the room. According to OSHA, maintaining a "safe room quality environment for classrooms" is required. (OSHA)

To address this, part of our solution involves creating a closure for these ports and ensuring the filters and ports can be efficiently opened and closed to maximize dust collection. The inefficiency of the collector—specifically its inability to work effectively when other machines are needed—directly impacts WHS students who are involved in and affected by the woodshop environment.

### Stakeholder Concerns and Connections (A3, A4)

**Primary Stakeholder #1-** Chris Delouchry Manufacturing Teacher who teaches Introduction and Advanced Manufacturing classes - Our main stakeholder who spends 100% of his day in the woodshop.

**Primary Stakeholder #2** - Students that take Manufacturing classes in the woodshop

**Secondary Stakeholders** - This group includes Walpole High School students and staff who are not directly present when the shop machines are operating but still require clean air and a healthy learning environment if dusty air and wood particles escape the woodshop room.

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### Problem Importance (A6)

This problem is very important as safety is always Walpole High School's number one priority. The first purpose of a dust collection system is safety. Even something as simple as reduced visibility due to contaminated air can lead to unsafe environments and accidents. A properly installed and maintained dust collection system can significantly increase woodshop safety, with statistics showing a 26% to over 90% reduction in harmful, airborne dust concentrations. As a result, this is a very significant problem as these systems are essential for preventing health issues and keeping the students in the workshop safe.+

Our Engineering Team will design an efficient, tight closure that blocks outside room air flow and improves the dust system's efficiency with quantifiable constraints such as costs less than \$15, weighs less than 5lbs. and has a measurement of 3 inch in radius.



**Figure 1 - Dowel Jig with Hose Connected**



**Figure 2 - Right open port**



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**Figure 3 - Ports Above Four Tables in WHS Woodshop.**

#### Works Cited

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<https://www.donaldson.com/en-us/industrial-dust-fume-mist/technical-articles/industrial-dust-collector-troubleshooting/#:~:text=Overloaded%20systems%20and%20blinded%20filters,up%20and%20ensure%20proper%20discharge.>

"Permissible Exposure Limits – OSHA Annotated Table Z-1 | Occupational Safety and Health Administration"

"Sanding Dust Collection Applications." *A.C.T. Dust Collectors*,

<https://www.actdustcollectors.com/applications/sanding-finishing.>

Figure 1- Shelton, M. Dowel Jig with Hose Connected. 13 February, 2026. Author's Personal Collection

Figure 2- Shelton, M. Right Port open. 13 February, 2026. Author's Personal Collection

Figure 3 - Shelton, M. Ports Above Four Tables in WHS Woodshop. 13 February, 2026. Author's Personal Collection